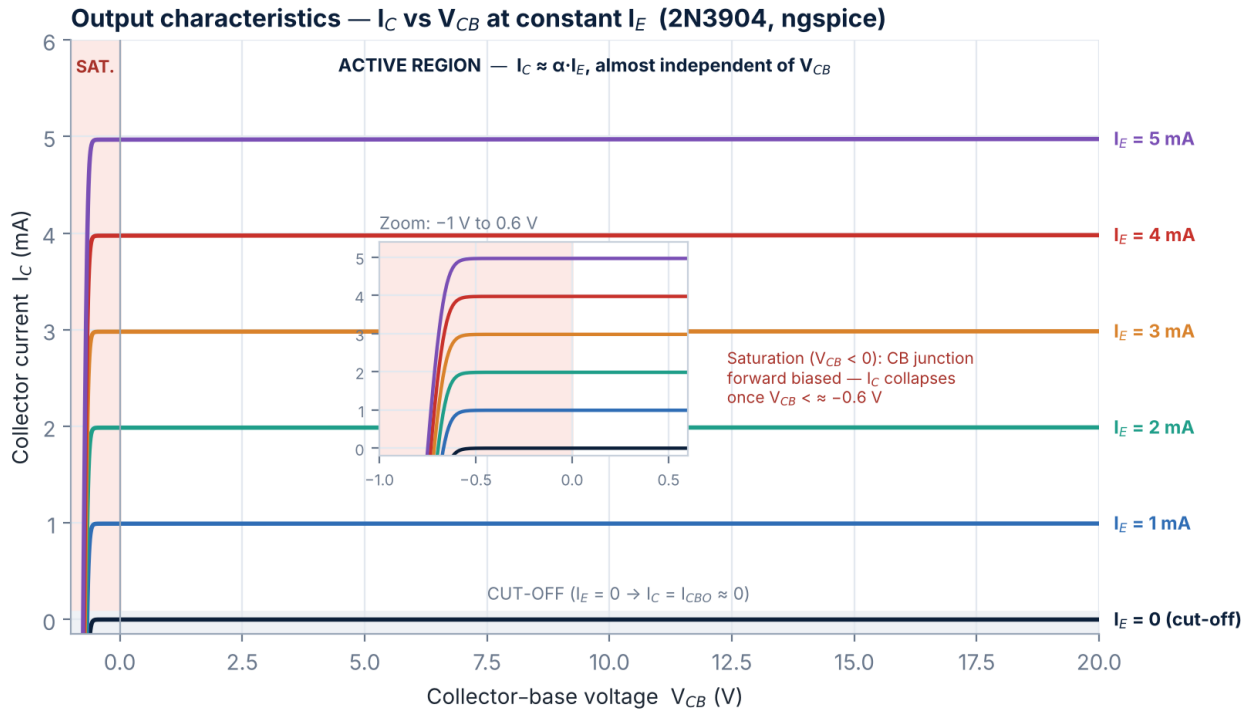
SIMULATION · OBSERVATION · ANALYSIS

BJT Common-Base Amplifier: Output Characteristics

Circuit simulation of an NPN BJT (2N3904) in common-base configuration in ngspice. Covers output and input characteristics, a working CB amplifier, and why the BJT is a **current-controlled current source** and why a **CB stage works as a current buffer**.



GROUP / BATCH

28

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GITHUB LINK

github.com/RehanBuildsWay/Group-Project-PartB

Folder "Project Part B" holds this document. Tap the link or scan the QR code.

1 Problem Statement & Objectives

Project Statements Part B — Question 2b

Q. NO.	PROBLEM DESCRIPTION
2b	Simulate, observe and analyze the output characteristics of a BJT common base amplifier, and explain the following: (i) BJT is called a current controlled current source (ii) CB-BJT can be used as current buffers

Group details

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Objectives

- **Simulate** the common-base (CB) configuration and plot the output characteristics I_C vs V_{CB} for constant I_E.
- **Observe** the active, saturation and cut-off regions; read α, output resistance and the input characteristic.
- **Analyze** a complete CB amplifier: Q-point, gain, input/output impedance, bandwidth and phase.
- **Explain**, with simulation evidence, (i) the BJT as a current-controlled current source and (ii) the CB stage as a current buffer.

Tools & device

Simulator	ngspice 42 (SPICE3-compatible): DC sweep, AC small-signal, transient and operating-point analyses
Transistor	2N3904 general-purpose NPN, standard Gummel-Poon SPICE model ($\beta_F \approx 416$ ideal, $V_{AF} = 74$ V)
Post-processing	Python (NumPy, Matplotlib); graphs are plotted straight from the simulator output data
Temperature	27 °C (SPICE default)

Contents

1	Problem statement & objectives	2	Theory of the common-base configuration
3	Simulation 1: characteristics test circuit	4	Output characteristics: results & observations
5	Input & transfer characteristics	6	Simulation 2: CB amplifier analysis
7	(i) BJT as a current-controlled current source	8	(ii) CB-BJT as a current buffer
9	Conclusion	10	Appendix: SPICE netlists

2 Theory — Common-Base Configuration

The base terminal is common to input and output

Configuration

In the common-base (CB) connection the **input is applied between emitter and base** and the **output is taken between collector and base**. The base is common to both loops (at ac ground in an amplifier).

- Input quantities: I_E, V_{EB}
- Output quantities: I_C, V_{CB}
- For normal (active) operation: E–B junction **forward** biased, C–B junction **reverse** biased.

Basic current relations

$$I_E = I_C + I_B$$
$$I_C = \alpha \cdot I_E + I_{CBO}$$
$$\alpha = I_C/I_E = \beta/(\beta+1) \quad (0.95 \ldots 0.995)$$
$$\beta = \alpha/(1-\alpha)$$

α (alpha) is the **common-base current gain**. It is slightly less than 1 because a small part of the emitter current recombines in the base and leaves as I_B.

Regions of the output characteristic

REGION	JUNCTION BIAS (E–B / C–B)	BEHAVIOUR ON THE I_C–V_{CB} GRAPH
Active	Forward / Reverse ($V_{CB} > 0$)	Flat, almost horizontal lines; $I_C \approx \alpha \cdot I_E$, nearly independent of V_{CB}. Used for amplification.
Saturation	Forward / Forward ($V_{CB} < 0$)	I_C falls sharply toward zero as V_{CB} goes about 0.5–0.7 V negative; the collector junction starts injecting carriers back.
Cut-off	Reverse / Reverse ($I_E = 0$)	Only the tiny leakage I_{CBO} flows (nA range); the bottom line lies on the axis.

Small-signal model (T-model)

$$r_e = V_T / I_E \approx 26\ \text{mV} / I_E$$
$$i_c = \alpha \cdot i_e \quad \text{(dependent current source)}$$
$$g_m = I_C / V_T$$

The collector is modelled as a **current source $\alpha \cdot i_e$** controlled by the emitter (input) current. This is the CCCS behaviour explained in Section 7.

Expected CB amplifier properties

Input impedance	**Very low** ≈ $r_e \parallel R_E$ (tens of Ω)
Output impedance	**High** ≈ R_C (device alone: MΩ)
Current gain	**α ≈ 1** (slightly < 1)
Voltage gain	**High** ≈ $g_m(R_C \parallel R_L)$
Phase shift	**0°** (non-inverting)
Bandwidth	**Wide** (no Miller multiplication of C_μ)

3 Simulation 1 — Characteristics Test Circuit

Circuit used to obtain the CB input and output characteristics

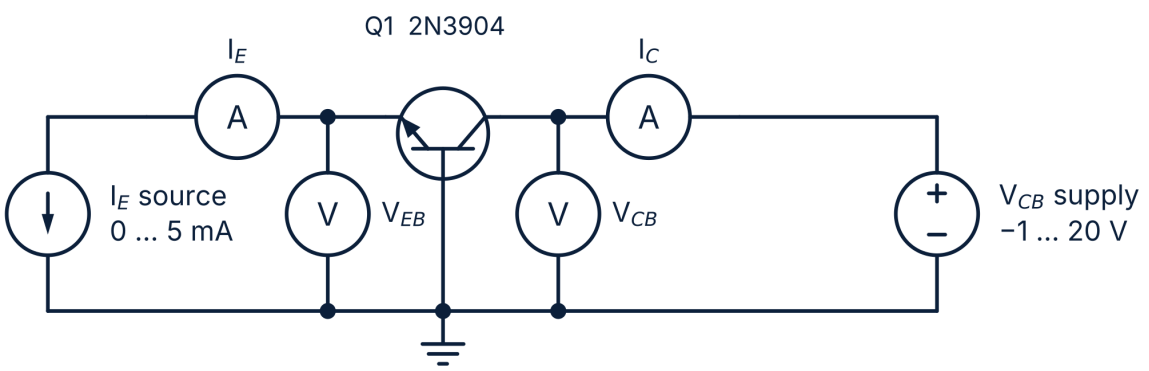


Figure 3.1 CB characteristics test circuit. The base is grounded (common), an adjustable current source sets I_E, and a variable supply sets V_{CB}. Ammeters/voltmeters show the quantities that SPICE records.

Procedure (as simulated)

1. **Output characteristics:** hold I_E constant (0, 1, 2, 3, 4, 5 mA) and sweep V_{CB} from −1 V to +20 V in 10 mV steps; record I_C.
2. **Input characteristics:** hold V_{CB} at 0, 5 and 10 V and sweep V_{EB} from 0 to 0.85 V; record I_E.
3. **Transfer characteristic:** hold V_{CB} = 5 V and sweep I_E from 0 to 10 mA; record I_C and I_B.

Why a current source on the emitter?

In CB the output characteristics are drawn for **constant input current I_E**. A SPICE current source holds I_E exactly constant while V_{CB} changes. On a lab bench the same effect comes from a V_{EE} supply with a large series resistor and an ammeter.

The ngspice `.dc` sweep with nested parameters produces the whole family of curves in one run (netlist in the Appendix).

4 Output Characteristics — Results & Observations

I_C vs V_{CB} for constant emitter current

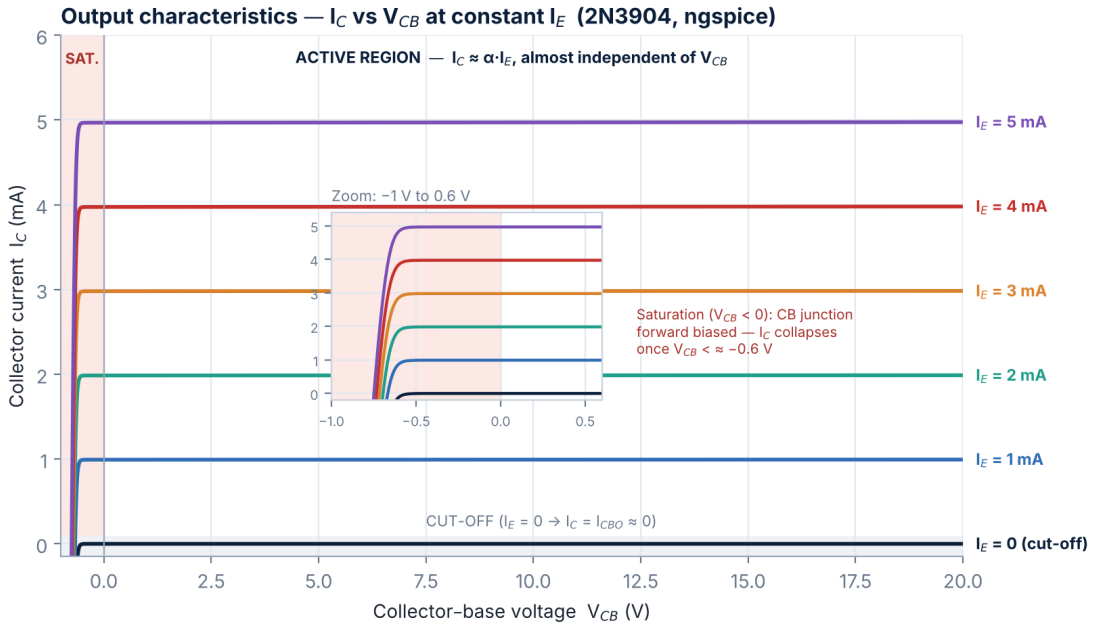


Figure 4.1 Simulated CB output characteristics of the 2N3904. The inset zooms into the saturation region near V_{CB} = 0.

I_E (mA)	I_C @ V_{CB} = 0 V	I_C @ 5 V	I_C @ 15 V	A = I_C/I_E (5 V)	OUTPUT RESISTANCE R_O
0	≈ 0	≈ 0 (I_{CBO})	≈ 0	—	—
1	0.9925 mA	0.9929 mA	0.9936 mA	0.9929	≈ 14.4 MΩ
2	1.9862 mA	1.9870 mA	1.9883 mA	0.9935	≈ 7.7 MΩ
3	2.9801 mA	2.9813 mA	2.9832 mA	0.9938	≈ 5.2 MΩ
4	3.9741 mA	3.9756 mA	3.9781 mA	0.9939	≈ 4.0 MΩ
5	4.9681 mA	4.9700 mA	4.9732 mA	0.9940	≈ 3.2 MΩ

$r_o = \Delta V_{CB} / \Delta I_C$ measured between V_{CB} = 2 V and 18 V from the simulation data.

Observations

- **Active region (V_{CB} > 0):** every curve is almost perfectly flat. Raising V_{CB} from 0 to 15 V changes I_C by only about 0.1 % (e.g. 0.9925 → 0.9936 mA at I_E = 1 mA).
- **$I_C \approx I_E$:** the collector current follows the emitter current with α ≈ 0.993–0.994, so the curves are equally spaced at ≈ 1 mA steps.
- **I_C flows even at V_{CB} = 0:** the built-in field of the collector junction still sweeps the carriers across. Current only collapses when the C–B junction is forward biased by ≈ 0.6–0.7 V (saturation).
- **Cut-off:** with I_E = 0 only the leakage I_{CBO} flows; it is practically zero in the simulation.
- **Very high output resistance** (several MΩ): the small upward slope comes from the Early effect (base-width modulation, V_{AF} = 74 V), and it is much weaker in CB than in CE.

5 Input & Transfer Characteristics

Supporting curves that complete the CB picture

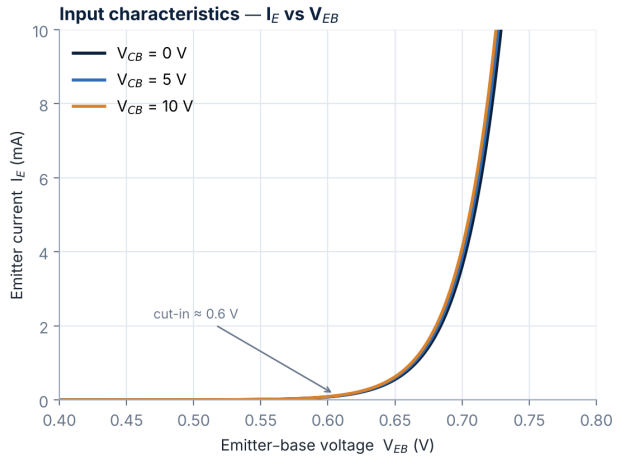


Figure 5.1 Input characteristics I_E vs V_{EB} at V_{CB} = 0, 5, 10 V.

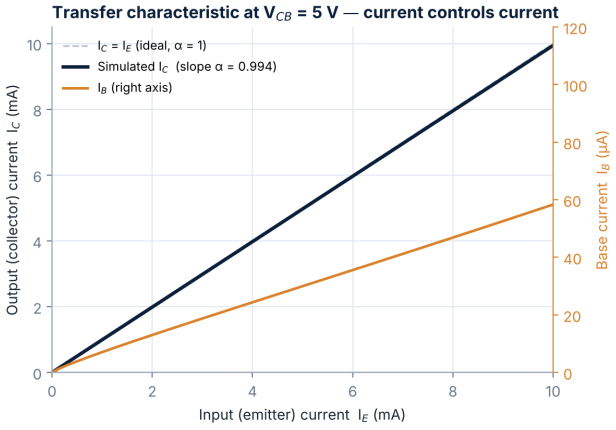


Figure 5.2 Transfer characteristic I_C vs I_E at V_{CB} = 5 V, with base current.

Input characteristic: observations

- Behaves like a forward-biased diode: cut-in ≈ **0.60 V**; I_E = 1 mA at **0.664 V** and 5 mA at **0.707 V** (V_{CB} = 5 V).
- Raising V_{CB} shifts the curve slightly to the left (≈ 2 mV per 5 V): the **Early effect** narrows the base.
- The steep slope means a **very low dynamic input resistance**, $r_e = \Delta V_{EB}/\Delta I_E \approx 26\text{ mV} / I_E$ (≈ 26 Ω at 1 mA).

Transfer characteristic: observations

- I_C is a **straight line** through the origin: linear fit gives **α = 0.9943**.
- Readings: I_E = 1 mA → I_C = 0.9929 mA; 5 mA → 4.970 mA; 10 mA → 9.942 mA.
- The difference is the base current: I_B ≈ 7.1 µA at 1 mA and 58 µA at 10 mA, i.e. β = α/(1−α) ≈ 140–170.

Key point

The output current is fixed by the **input current** (linear, α ≈ 1) and is almost unaffected by the **output voltage**. These two facts are the basis of both explanations in Sections 7 and 8.

6 Simulation 2 — CB Amplifier

A practical voltage-divider-biased common-base amplifier

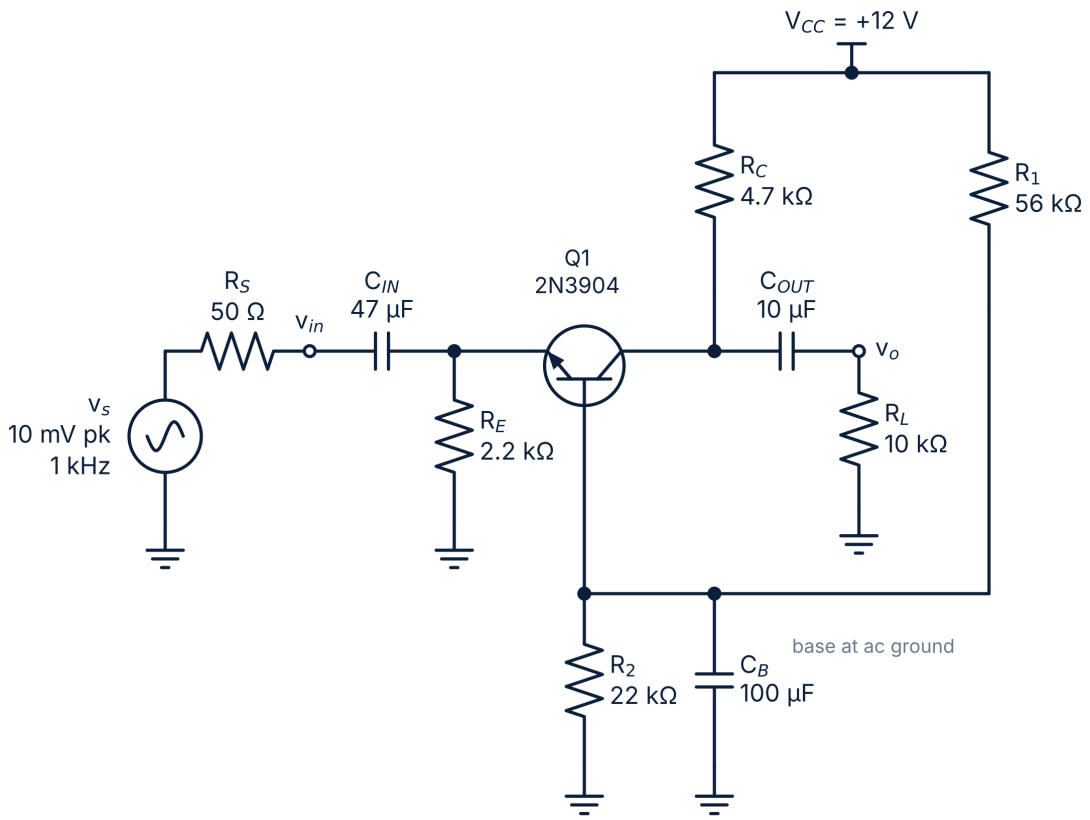


Figure 6.1 CB amplifier. R_1/R_2 bias the base and C_B puts it at ac ground. The signal enters at the emitter through C_{IN}, and the output is taken from the collector through C_{OUT}.

DC operating point (Q-point): ngspice .op

V_B / V_E / V_C	**3.25 V / 2.58 V / 6.52 V**
I_C / I_B	**1.167 mA / 8.30 µA**
$V_{CB} = V_C - V_B$	**3.26 V** → C–B reverse biased ACTIVE
$\beta = I_C/I_B$; $\alpha = \beta/(\beta+1)$	**140.6 ; 0.9929**
g_m ; $r_e = \alpha/g_m$	**44.4 mS ; 22.4 Ω**

Performance: simulated vs. hand calculation

PARAMETER	THEORY	SIMULATED
Input impedance Z_{in}	$r_e \| R_E \approx 22.2$ Ω	**23.5 Ω**
Output impedance Z_{out}	$\approx R_C = 4.7$ kΩ	**4.60 kΩ**
Voltage gain A_v	$g_m(R_C \| R_L) \approx 142$	**133.6 (42.5 dB)**
Current gain i_c/i_e	$\alpha \approx 0.993$	**0.993**
Phase shift	0°	**0° (in phase)**
Bandwidth (−3 dB)	wide	**145 Hz – 16.6 MHz**

A_v is a little below the ideal value because the model's base resistance (r_b) and the Early effect are not in the simple hand formula. The load current is $i_L = \alpha \cdot i_e \cdot R_C/(R_C+R_L) = 0.31 \cdot i_e$, since R_C shares the ac collector current with R_L.

6 CB Amplifier — Waveforms & Frequency Response

Transient (time-domain) and AC (frequency-domain) analysis

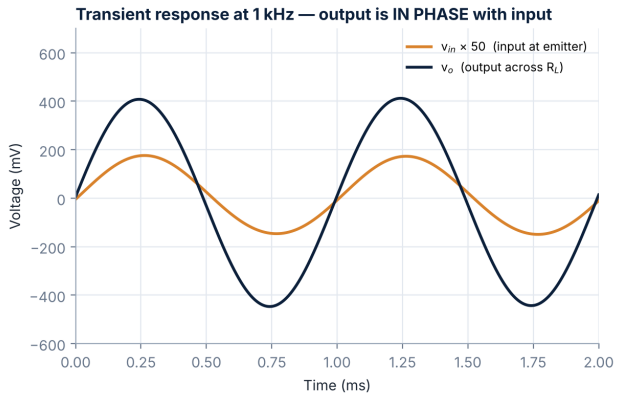


Figure 6.2 Transient at 1 kHz: input at the emitter (×50) and output across R_L.

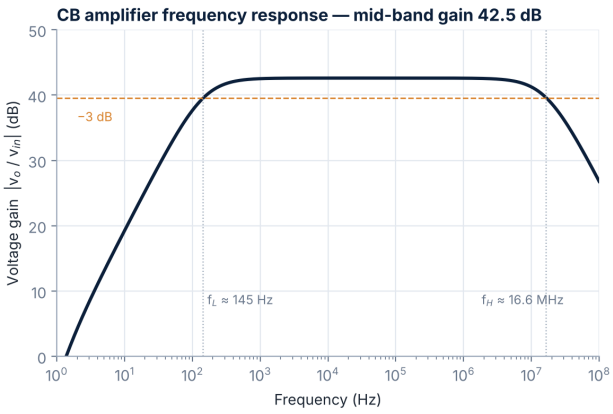


Figure 6.3 Frequency response: mid-band gain 42.5 dB, −3 dB points at 145 Hz and 16.6 MHz.

Analysis of the amplifier results

- **Non-inverting:** output and input peaks line up (Fig. 6.2). Increasing the emitter voltage reduces V_{BE} and I_C, so the collector voltage *rises*; the CB stage has no phase inversion.
- **High voltage gain but unity current gain:** a 6.5 mV_{pp} input gives a 0.86 V_{pp} output ($A_v \approx 133$), while the collector current equals α × the emitter current. All the voltage gain comes from transferring the same current from a **low impedance (≈ 23 Ω)** to a **high impedance (≈ 4.6 kΩ)**.
- **Low input impedance:** $Z_{in} \approx r_e \approx 23\ \Omega$. A voltage source with even 50 Ω internal resistance loses about two-thirds of its signal (10 mV → 3.2 mV), which is why CB is seldom used as a voltage-input stage. It is ideal, however, for a **current-signal** input (Section 8).
- **Wide bandwidth:** the base is grounded, so the collector–base capacitance C_μ is **not** Miller-multiplied as it is in CE. The upper cut-off reaches ≈ 16.6 MHz. The lower cut-off (145 Hz) is set mainly by C_{IN} working into the very low Z_{in}.

7 (i) Why a BJT is Called a Current-Controlled Current Source

Explanation supported by the simulated characteristics

In one sentence

The BJT's **output (collector) current is set by its input current** (I_E or I_B) and stays **almost constant whatever the output voltage**. That is exactly the definition of a **current-controlled current source (CCCS)**.

1 · Controlled by current

$$I_C = \alpha \cdot I_E \quad \text{(CB)}$$
$$I_C = \beta \cdot I_B \quad \text{(CE)}$$

In the transfer curve (Fig. 5.2) I_C is a straight line through the origin with slope $\alpha = 0.994$. Doubling I_E doubles I_C (1 → 0.993 mA, 2 → 1.987 mA, 5 → 4.970 mA). The **control variable is a current**, not a voltage.

2 · Acts as a current source

An ideal current source delivers a fixed current regardless of the voltage across it, so its I–V line is **horizontal**.

The simulated output curves (Fig. 4.1) are horizontal in the active region: from V_{CB} = 0 to 15 V, I_C changes by only ≈ 0.1 %, giving an output resistance of **3–14 MΩ**. The collector therefore behaves as a near-ideal current source.

3 · Physical reason

- The forward-biased E–B junction **injects** electrons into the thin base. The amount injected is fixed by I_E.
- About 99.3 % diffuse across the base and are **swept** into the collector by the reverse-biased C–B junction field. The rest recombine and form I_B.
- Making V_{CB} larger only strengthens a field that already collects almost every carrier, so I_C barely changes (only the small Early effect).

4 · Circuit model

That is why every small-signal BJT model contains a **dependent current source**:

$$\text{T-model:} \quad i_c = \alpha \cdot i_e$$
$$\text{h-model:} \quad i_c = h_{fe} \cdot i_b$$

In circuit theory this element, a current source whose value is a constant × another branch current, is the **CCCS**. (A FET, controlled by gate voltage, is a **VCCS** instead.)

Evidence summary from our simulation

CCCS PROPERTY	WHAT WE OBSERVED	WHERE
Output current ∝ input current	I_C/I_E = 0.993–0.994 for 1–10 mA (linear)	Fig. 5.2, Table §4
Output current independent of output voltage	< 0.15 % change in I_C for 0 → 15 V	Fig. 4.1
Very high output resistance	r_o ≈ 3.2–14 MΩ	Table §4
No output current without input current	I_E = 0 → $I_C \approx I_{CBO} \approx 0$	Fig. 4.1 (cut-off)

8 (ii) Why a CB-BJT Can Be Used as a Current Buffer

Explanation plus a dedicated buffer simulation

In one sentence

A current buffer must **take in all the current a source gives** (very low input impedance), **pass it on unchanged** (current gain ≈ 1) and **deliver it to any load** (very high output impedance). The CB stage has all three properties, so it is the BJT's natural **current buffer (current follower)**, just as the common-collector stage is a voltage buffer.

IDEAL CURRENT BUFFER NEEDS	CB-BJT PROVIDES	SIMULATED VALUE
$Z_{in} \to 0$, so the source's current is not lost in its own resistance	$Z_{in} \approx r_e = V_T/I_E$	**≈ 23 Ω**
Current gain = 1	$A_i = \alpha$	**0.993**
$Z_{out} \to \infty$, so load changes do not change the current	Collector is a CCCS (Section 7)	**MΩ (device)**
Isolation between source and load	Base grounded; output barely affects input (low reverse transfer)	—

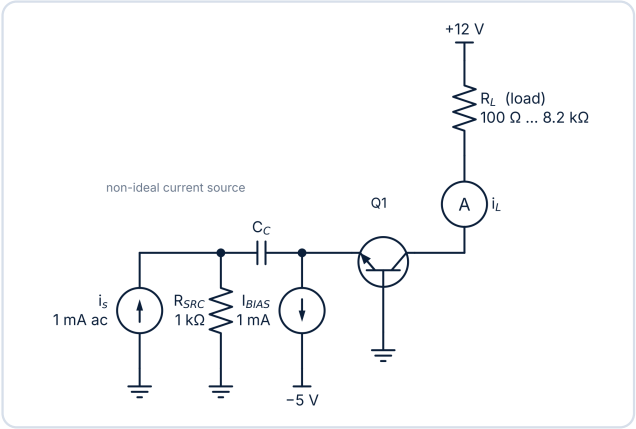


Figure 8.1 Buffer test: a 1 mA ac source with a 1 kΩ internal resistance drives the emitter. R_L is varied from 100 Ω to 8.2 kΩ.

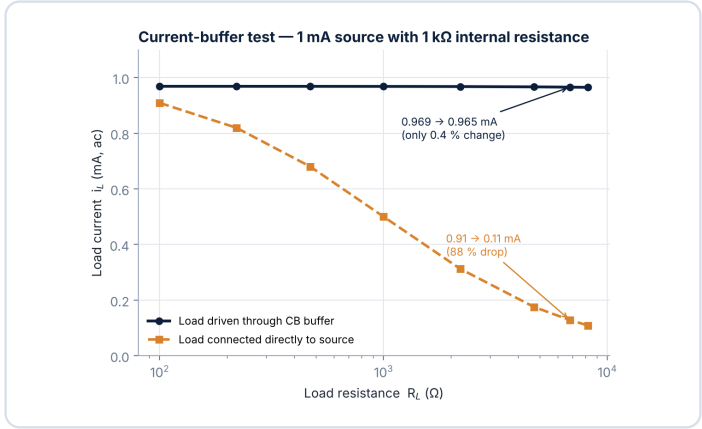


Figure 8.2 Load current vs load resistance: through the CB buffer (navy) vs connected directly to the source (orange).

Buffer simulation result

R_L	DIRECT (MA)	VIA CB (MA)
100 Ω	0.909	**0.969**
1 kΩ	0.500	**0.968**
4.7 kΩ	0.175	**0.967**
8.2 kΩ	0.109	**0.965**

Analysis

- **Without a buffer** the source current divides between its own 1 kΩ and the load, so i_L drops by **88 %** as R_L rises from 100 Ω to 8.2 kΩ.
- **With the CB buffer** the 23 Ω input takes ≈ 97.5 % of the source current, and the collector delivers α × that to the load: i_L stays at **0.965–0.969 mA**, a change of only **0.4 %** over an 82× change in load.
- The load is **isolated** from the source: the source "sees" 23 Ω whatever R_L is.

9 Conclusion

Summary of observations and analysis

α = 0.994	3–14 MΩ	≈ 23 Ω	42.5 dB	16.6 MHz
CB CURRENT GAIN	OUTPUT RESISTANCE	INPUT IMPEDANCE	VOLTAGE GAIN (A_V ≈ 134)	UPPER CUT-OFF

- The CB output characteristics of the 2N3904 were simulated in ngspice for I_E = 0–5 mA. They show a **flat active region** ($I_C \approx \alpha \cdot I_E$, α ≈ 0.993–0.994), a **saturation region** for V_{CB} below ≈ −0.6 V, and a **cut-off region** at $I_E = 0$.
- The input characteristic is a forward-biased diode curve (cut-in ≈ 0.6 V) with a very small dynamic resistance; the transfer characteristic is linear with slope α.
- A voltage-divider-biased CB amplifier was analyzed: Q-point I_C = 1.17 mA, V_{CB} = 3.26 V (active); **A_v ≈ 134 in phase**, **$A_i \approx \alpha$**, **Z_{in} ≈ 23 Ω**, **Z_{out} ≈ 4.6 kΩ**, bandwidth 145 Hz – 16.6 MHz.
- **(i)** The BJT is a **current-controlled current source** because its collector current is proportional to the input current ($I_C = \alpha \cdot I_E = \beta \cdot I_B$) and nearly independent of the collector voltage; the simulation shows MΩ output resistance.
- **(ii)** The CB-BJT is a **current buffer** because it combines very low input impedance, unity current gain and very high output impedance. In simulation it held the load current within 0.4 % while R_L varied 82×, compared with an 88 % drop without it.

Where CB current buffers are used

Cascode amplifier (CE + CB: the CB buffers the CE collector current, removing the Miller effect and raising bandwidth) · **photodiode / sensor front-ends** (low-impedance input for a current-output sensor) · **high-output-impedance current mirrors and current sources** · **RF/VHF amplifiers** (wide bandwidth, good input–output isolation).

Learning outcome

The CB configuration does not amplify current. It **moves** a current from a low-impedance node to a high-impedance node. This gives a large voltage gain, excellent bandwidth and good isolation, and it makes the CB stage a key building block in cascode amplifiers, current mirrors and RF front-ends.

10 Appendix — SPICE Netlists

ngspice 42 input files used for every result in this report

```
* ---- Device model (2N3904, standard Gummel-Poon)
.model 2N3904 NPN(IS=6.734f XTI=3 EG=1.11 VAF=74.03 BF=416.4 NE=1.259 ISE=6.734f
+ IKF=66.78m XTB=1.5 BR=.7371 NC=2 ISC=0 IKR=0 RC=1 CJC=3.638p MJC=.3085 VJC=.75
+ FC=.5 CJE=4.493p MJE=.2593 VJE=.75 TR=239.5n TF=301.2p ITF=.4 VTF=4 XTF=2 RB=10)
```

```
* ---- A. CB output characteristics
Q1  c 0 e 2N3904      ; base grounded
IE  e 0 1m            ; constant emitter current
VCB c 0 0
.control
foreach ie 0 1m 2m 3m 4m 5m
  alter IE dc = $ie
  dc VCB -1 20 0.01
  let ic = -i(VCB)
end
.endc
```

```
* ---- B. CB input characteristics
Q1  c 0 e 2N3904
VEB 0 e 0             ; emitter driven negative
VCB c 0 0
.control
foreach v 0 5 10
  alter VCB dc = $v
  dc VEB 0 0.85 0.001
end
.endc
* ---- C. Transfer (VCB = 5 V)
dc IE 0 10m 0.1m
```

```
* ---- D. CB amplifier
VCC vcc 0 12
R1  vcc b 56k
R2  b   0 22k
CB  b   0 100u        ; base at ac ground
RE  e   0 2.2k
RC  vcc c 4.7k
Q1  c b e 2N3904
VS  s 0 DC 0 AC 1 SIN(0 10m 1k)
RS  s sx 50
CIN sx e 47u
COUT c o 10u
RL  o 0 10k
.op
.ac dec 50 1 100Meg
.tran 2u 3m 1m
```

```
* ---- E. Current-buffer test
VCC   vcc 0 12
RL    vcc c {100 … 8.2k}
Q1    c 0 e 2N3904
IBIAS e nb 1m         ; DC bias 1 mA
VNEG  nb 0 -5
ISIG  0 e2 AC 1m      ; 1 mA ac signal
RSRC  e2 0 1k         ; source resistance
CC    e2 e 1000u
.ac lin 1 1k 1k
* compare: iL = 1m*1k/(1k+RL) direct
```

Batch 28 · Mohd Rehan Hasan · Kanika Sahlot · Pratishtha Dhar · Yashveer Sharma · github.com/RehanBuildsWay/Group-Project-PartB